Towards a replication culture in phonetic research: Speech production research in the classroom

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Abstract

Our understanding of human sound systems is increasingly shaped by experimental studies. What we can learn from a single study, however, is limited. It is of critical importance to evaluate and substantiate existing findings in the literature by directly replicating published studies. Our publication system, however, does not reward direct replications in the same way as it rewards novel discoveries. Consequently, there is a lack of incentives for researchers to spend resources on conducting replication studies, a situation that is particularly true for speech production experiments which often require resourceful data collection procedures and recording environments. In order to sidestep this issue, we propose to run direct replication studies with our students in the classroom. This proposal offers an easy and cheap way to conduct high-powered replication studies and has valuable pedagogical advantages for our students. As an existence proof, we report on two such replication studies in the classroom on incomplete neutralization, a speech phenomenon that has sparked many methodological debates in the past. We show that in our classroom replication studies, we not only replicated incomplete neutralization effects, but the studies yielded effect magnitudes comparable to laboratory experiments and meta analytical estimates. We conclude by discussing potential challenges to this approach and discuss possible ways forward that can help us substantiate our scientific record.

Keywords

Final devoicing, incomplete neutralization, German, replication, pedagogy

1 Introduction

Our understanding of human communication and its cognitive underpinnings is increasingly shaped by experimental data. For example, experimental work has repeatedly demonstrated that putative neutralizations of contrasting speech sounds are incomplete, i.e. there are small but systematic acoustic and articulatory differences between neutralized and non-neutralized speech sounds (e.g. Mitleb, 1981; Roettger et al., 2014). These findings led to important discussions about cognitive representations of speech sounds (e.g. Dinnsen & Charles-Luce, 1984; Charles-Luce, 1985; Port & O'Dell, 1985; Oostendorp, 2008) and served as a springboard for theories about the organization of lexical representations (e.g., Ernestus & Baayen, 2006; Goldrick, Folk, & Rapp, 2010; Winter & Roettger, 2011; Seyfarth et al., 2018).

With an increasingly growing body of experimental evidence, it is of critical importance to evaluate and substantiate existing findings in the literature. This is particularly important because the evidence provided by a single study is limited to the concrete research method and context. Results may depend on idiosyncratic properties of the sample, random measurement fluctuations, or human error due to variation in procedures and the experimenters involved.

In order to ensure the reliability of scientific findings, we should thus directly replicate our findings. However, running experiments is expensive and time-consuming. This is particularly true for speech production experiments which often require specific data collection procedures and recording environments. Moreover, our current publication system does not reward direct replications (Koole & Lakens, 2012; Makel, Plucker, & Hegarty, 2012; Nosek, Spies, & Motyl, 2012). This lack of incentives often discourages researchers from investing time and money in direct replications.

The present paper discusses a possible solution to this dilemma. In line with several developments throughout the social sciences, we propose making replication studies an explicit part of the linguistic curriculum. In other words, while we are teaching our students about experimental methods, we should involve them in running high-powered replication studies of recent published findings. While this proposal has been suggested for the psychological sciences (Frank & Saxe, 2012; Grahe et al., 2014, 2016; Hawkins et al., 2018), we believe it is necessary to re-evaluate these suggestions in light of our own empirical practices. Phonological research is often informed by speech production experiments that exhibit their own unique methodological challenges. For the present purpose, we will reiterate these points with an accessible example from phonology.

The remainder of this paper is organized as follows: In §2 we will discuss the crisis of confidence within quantitative social sciences, which was mainly triggered by the failure to replicate 'established' findings (§2.1). Despite the importance of replication studies, only very few of them are conducted. We discuss reasons why researchers are reluctant to run replication studies and discuss a practical solution: running replication studies in the classroom (§2.2). To demonstrate the feasibility of this approach for speech production studies, we discuss incomplete neutralization of German final devoicing as a test case (§2.3). In §3, we describe two experimental studies, replicating Roettger et al.'s findings (2014), the as of yet largest lab-based experiment on incomplete neutralization in German. In §4, we present the results, and compare them to the results of the original study. We conclude in §5 that despite the level of control in the lab (and in turn the lack of control in our study), high-powered speech production studies executed by students can yield valuable evidence that contributes to the evidence accumulation within our field. It is resource-economical and offers feasible ways to replicate empirical evidence from the literature and, at the same time, it substantiates the external validity of these findings.

2 Background

2.1 The replication crisis and why we should be worried

The credibility of scientific findings is mainly rooted in the evidence supporting it. We assess the validity of this evidence by constantly reviewing and revising our methodologies, and by extending and replicating findings. We end up having to *trust* the authors. Trust, however, is not what science is built on. The oldest continuously operating scientific society in the world, the Royal Society subscribed in 1660 to the motto *nullius in verba*, roughly meaning "take nobody's word for it."

The replication of a finding has been an essential component of the scientific method (e.g. Dunlap, 1925; Campbell, 1969; Zwaan et al., 2018). Only by replicating empirical results and evaluating the accumulated evidence can we gain a better understanding of the world. In recent coordinated efforts to replicate published results, the social sciences uncovered unexpectedly low replicability rates, a state of affairs that has been coined the 'replication crisis'.

This crisis spans some of the most cited studies in the social sciences. For example, Carney et al. (2010) presented data suggesting that after adopting high power poses (either having your hands behind your head, your legs on the desk, or your feet apart leaning on a desk), subjects were more likely to make riskier bets and exhibited measurably higher testosterone and lower cortisol levels, compared with subject who adopted low power poses. Carney et al. (2010) has been cited over 800 times, has sparked many follow-up studies, and resulted in a highly visible TED talk by Amy Cuddy, making their findings 'public knowledge'. However, in a large-scale replication attempt with more than four times as many subjects, Ranehill et al. (2015) could neither find an effect of power posing on hormones nor on risk tolerance.

Another well-known example is the finding that smiling putatively makes you happy: Strack, Martin and Stepper (1988) published a seminal study, suggesting that subjects who hold a pen between their teeth, forcing them to smile, rate cartoons as funnier when compared to them holding a pen between their lips, forcing them to pout. The finding appeared to be consistent with the idea that our facial expressions are not just a reflection of our emotions but can also cause them. This paper has been cited over 1800 times, again having become textbook knowledge. Wagenmakers et al. (2016) attempted to replicate the original results, involving 17 independent labs and nearly 2000 subjects. They found overall no relationship between mouth configuration and subject's emotional state.

As these selective examples show, seemingly unambiguous evidence provided by a single study can be falsely held to be generalizable. When undetected, such misinterpretations can have practical consequences, potentially leading to theoretical claims which themselves may misguide future research.

The Open Science Collaboration (2015) tried to replicate 100 studies that were published in three high-ranking psychology journals, assessing whether the replications and the original experiments yielded the same result. A widely shared summary of their project was that only 36% of the attempted replications were successful, i.e. finding a significant effect with the same directionality as the original study.

Concerns about the replicability of findings have been raised for many other disciplines including the medical sciences (e.g. Ioannidis, 2005), cancer research (Errington et al., 2014), the neurosciences (Wager et al., 2009), economics (Camerer et al., 2016), and genetics (Hewitt, 2012).

Most importantly, it is a very real problem for quantitative linguistics, too. For example, Stack, James, and Watson (2018) recently failed to replicate a widely cited effect of syntactic adaptation by Fine, Jaeger, Farmer, and Qian (2013). After failing to find the original effect in a conceptual replication, they went back and directly replicated the original study with appropriate statistical power. They found no evidence for syntactic adaptation as reported by the original study. Nieuwland et al. (2018) recently tried to replicate a seminal study by DeLong et al. (2005) which is considered a landmark study for the predictive processing literature and which has been cited over 800 times. In their preregistered multi-site replication attempt (9 laboratories, 334 subjects), Nieuwland et al. were not able to replicate some of the key findings of the original study. Now turning our attention to quantitative linguistics, it seems we already are facing the beginning of our very own crisis of confidence.

There are many possible reasons why a study cannot be replicated (e.g., Ioannidis, 2005; Simmons et al., 2011; Gelman & Loken, 2014; Silberzahn et al., 2017). Idiosyncratic properties of the sample, measurement errors, and the human factor in data collection and analysis can all have a noticeable influence on the interpretation of experimental results. Moreover, statistical issues related to low power (e.g. for recent discussion of power in speech research see Kirby & Sonderegger, 2018; Nicenboim et al., 2018), violation of the independence assumption (Nicenboim & Vasishth, 2016; Winter, 2011, 2015), exploitation of researcher degrees of freedom (Roettger, 2018) and the 'significance' filter (i.e. treating results publishable because $p < 0.05$ leads to over optimistic expectations of replicability, see Vasishth et al., 2018) can lead to skewed results, which, at face, present themselves as unambiguous. Therefore, it would come as no surprise to us if there are a large number of experimental linguistic findings that may not stand the test of time.

2.2 Why don't we replicate more?

The above replication failures sparked a tremendously productive discourse throughout the quantitative sciences and led to quick methodological advancements and best practice recommendations. For example, there are several coordinated efforts to directly replicate important findings by multi-site projects such as the ManyBabies project (Frank et al., 2017) and Registered Replication Reports (Simons, Holcombe, &, Spellman, 2014). These coordinated efforts can help us put theoretical foundations on a firmer footing. However, the logistic and monetary resources associated with such large-scale projects are not always accessible for everyone in the field.

Importantly, the popularity of replication studies is even further diminished because the time and resource investment that are necessary for such studies are not appropriately rewarded in contemporary academic incentive systems (Koole & Lakens, 2012; Makel, Plucker, & Hegarty, 2012; Nosek, Spies, & Motyl, 2012). Both successful replications (Madden, Easley, & Dunn, 1995) and repeated failures to replicate (e.g., Doyen, Klein, Pichon, & Cleeremans, 2012) are only rarely published and if they are, it is usually in a less prestigious outlet than the original findings. An additional factor in this cost-benefit-equation is the often-cited preference for laboratory experiments (Xu, 2010 for a discussion of this argument in the context of speech sciences): There is a common belief that even relatively simple acoustic recordings with participant populations that are easy to recruit must be conducted under extremely well controlled laboratory conditions (but see Wagner et al., 2015 and references therein).

This concern is particularly relevant for the majority of speech production experiments, which necessitates at least a recording device and ideally a sufficiently quiet recording environment. These requirements make data collection often resource-intensive (at least in comparison to survey-based research in the social sciences). Usually, speakers participate in logistically costly laboratory experiments and factors related to the testing environment are controlled for by keeping them constant. For example, we control the recording environment by recording all speakers under sufficiently similar conditions, including conducting the experiment in the same room, by the same experimenter, with the same recording equipment, etc. This obviously poses pragmatic limits on data collection efficiency, feeding into potential reluctance of conducting replication studies. The tension between aiming for a controlled experimental environment and minimizing resource cost is a serious dilemma.

Commonly this dilemma is solved by testing only a small number of speakers. For example, Ladefoged (2003) recommends six speakers as a sufficient sample for speech production studies. While certainly not intended as a statistical argument, this reflects a common misconception of experimental work in quantitative linguistics. Published studies in linguistics and related areas often have very low statistical power, i.e., low probability that the statistical test will reject a false null hypothesis. For example, Kirby and Sonderegger (2018) report simulation studies showing that speech production studies using six speakers may have as low power as 6% (dependent on the effect size), i.e. having a 6% probability that the statistical test will reject a false null hypothesis. While typical subject numbers differ across subdomains of quantitative linguistics, six participants is not an uncommon sample size in phonetic experiments (see Roettger & Gordon, 2017, on the word stress literature; and Nicenboim et al., 2018, for a more general discussion).

It is important to note that we are neither bound to laboratory experiments nor to small sample sizes. As opposed to rigid laboratory settings, there are alternative data collection approaches that are relatively resource-conserving. These empirical approaches often introduce a less controlled experimental environment, but allow to gather large data sets quickly and efficiently. Trading off control against data collection convenience and sample size is in line with a noticeable trend towards using crowdsourcing platforms such as Amazon Mechanical Turk (e.g. Mason & Suri, 2012). Crowdsourced experimental data has been repeatedly cross-validated with laboratory experiments (Behrend, Sharek, Meade, & Wiebe, 2011; Goodman, Cryder, & Cheema, 2012; Shapiro, Chandler, & Mueller, 2013) and can thus be considered a valid alternative to laboratory experiments.

However, not all types of empirical investigations can be outsourced to browser-based applications, at least not yet. Speech production studies still necessitate a recording device and (depending on the measure) a relatively quiet recording environment. Even though we are optimistic that we can outsource many empirical questions to crowdsourcing platforms soon, conducting high powered replication studies on platforms such as Amazon Mechanical Turk still costs a substantial amount of funding, making it, again, less attractive for many researchers.

We would like to put forward an alternative strategy. We propose to run high-powered replication studies conducted by students as class assignments (Frank & Saxe, 2012; Grahe et al., 2014, 2016; Hawkins et al., 2018). In other words, while we are teaching our students about experimental methods, we ask them to run high-powered replication studies of published findings. Letting students run these studies as class assignments can be a feasible and affordable way to replicate often-cited experimental findings.

Not only would this amplify the number of independent replications, enabling us to put our theories on a firmer footing, but this model would also create a educationally rewarding environment for the students to learn about the basic tenets of science (Frank & Saxe, 2012). While some might be concerned that speech production experiments are not suited for this proposal, this paper will provide an existence proof that the apparent noisiness of speech production data can be overcome by large sample sizes, revealing comparable population estimates and, crucially, higher precision (i.e. uncertainty about our estimate of interest, see Vasishth et al., 2018) than controlled laboratory studies. We discuss two production experiments that were run in the classroom, investigating a notoriously brittle speech production effect that has caused many methodological and theoretical debates within the last 40 years or so: Incomplete neutralization of final devoicing.

2.3 A case study: Incomplete neutralization

Within the last four decades, incomplete neutralization has proven to be a fruitful ground for methodological debates that advanced methodological rigor and the critical assessment of empirical findings within the phonetic sciences (e.g. Manaster-Ramer, 1996; Winter & Roettger, 2011; Roettger et al., 2014). Recently, incomplete neutralization has served as a prime example to facilitate discussions about statistical concepts such as Type-I and Type-II errors, pseudoreplication, and meta-analysis (Roettger et al., 2014; Kirby & Sonderegger, 2018; Nicenboim et al., 2018).

Incomplete neutralization refers to the finding that an assumed phonological neutralization such as final devoicing is phonetically not complete. Cross-linguistically, final devoicing is a wide-spread phonological alternation. The textbook example is the German Auslautverhärtung: German contrasts voiced obstruents intervocalically but neutralizes the contrast syllable or word finally in favor of voiceless obstruents (cf. 1-2):

(1) /d/ in syllable final position: Rad [Raːt] 'wheel'
/d/ in syllable initial position: Räder [Rɛːdɐ] 'wheels'

(2) /t/ in syllable final position: Rat [Raːt] 'council'
/t/ in syllable initial position: Räte [Rɛːtə] 'councils'

In intervocalic position, the voicing contrast of oral stops can be manifested by different acoustic dimensions, such as the preceding vowel duration, glottal pulsing during the closure, closure duration, or voice onset time (among other cues, e.g. Lisker, 1986), with voiced stops exhibiting longer preceding vowels, more glottal pulsing during the closure, a shorter closure duration, and shorter (or negative) voice onset time. These acoustic differences are supposedly neutralized in syllable final position, implying that the acoustic form of the alveolar stop in *Rad* is identical to the alveolar stop in *Rat*.

This argument was grounded on ear-phonetic assessments of traditional linguistic descriptions (e.g. Jespersen, 1920; Trubeckoj, 1939; Wiese, 1996). However, as soon as linguists started testing this assumption experimentally, small but consistent acoustic and/or articulatory differences between words such as *Rad* and Rat were found, suggesting that in German, this neutralization can be considered incomplete (Dinnsen & Garcia-Zamor, 1971; Taylor, 1975; Mitleb, 1981; Port & O'Dell, 1985; Charles-Luce, 1985; Port & Crawford, 1989; Roettger et al., 2014). While the direction of the difference resembles the non-neutralized contrast (Räder vs. Räte), the magnitude of the difference turned out to be much smaller. For example, in intervocalic position, voiced stops are characterized by large durational differences in the preceding vowel (the /ɛː/ in Räder vs. the /ɛː/ in Räte) with reports on vowel duration differences ranging from 20 to 40 ms (see Mitleb, 1981; Fuchs, 2005; Roettger et al., 2014). In syllable final position, however (i.e. the putative neutralized position), this difference has been reported to be much smaller. For example, Roettger et al. (2014) report a vowel duration difference of approximately 9 ms between devoiced and voiceless stops in German. These effects, albeit repeatedly found, are most likely below the just noticeable difference threshold (e.g. Kohler, 2012) and might not be perceivable by listeners in isolation. This is in line with listeners' poor identification performances in forced-choice tasks (e.g. Roettger et al., 2014), suggesting a very limited functionality value of these subtle acoustic differences.

The main response to these findings was acknowledgement of the evidence for incomplete neutralization, leading to attempts to implement this phenomenon into formal models of phonological representations (e.g. Dinnsen & Charles-Luce, 1984; Charles-Luce, 1985; Port & O'Dell, 1985; van Oostendorp, 2008). More recent accounts to incomplete neutralization are rooted in psycholinguistic models of lexical organization, suggesting that incomplete neutralization and similar phenomena are artifacts of lexical co-activation (e.g., Ernestus & Baayen, 2006; Goldrick, Folk, & Rapp, 2010; Kleber et al., 2010; Winter & Roettger, 2011; Roettger et al., 2014; Seyfarth et al., 2018).

In contrast, there was also a substantial number of researchers that have remained skeptical regarding incomplete neutralization, mainly worrying that incomplete neutralization could be a methodological artifact (Manaster-Ramer, 1996; Kohler, 2007, 2012; Roettger et al., 2014). Some studies even failed to replicate incomplete neutralization (Fourakis & Iverson, 1984; Inozuka, 1991; Jessen & Ringen, 2002; Piroth & Janker, 2004, but see Nicenboim et al., 2018, arguing that most of these findings are due to too little power).

One camp of scholars interprets the available evidence in favor of incomplete neutralization, with important implications for theories of speech production and cognitive representations, while others interpret the available evidence as either insufficient or pointing towards incomplete neutralization being a methodological artifact. The latter position has led to productive methodological debates, drawing attention to the problem of how much a single study can really tell us about the world. Nicenboim et al. (2018) took this as a point of departure and gathered all eligible studies on German incomplete neutralization to quantify the overarching evidence and the uncertainty it is associated with. Using a Bayesian multilevel meta-analysis, they showed that across eligible 14 studies, there was substantial evidence for an incomplete neutralization effect. The meta-analytical estimate for the magnitude of the effect was 10 ms (95% credible interval: 6-16 ms).

Despite its presumed brittleness, incomplete neutralization of final devoicing appears to become more and more substantiated as evidence accumulates. It arguably becomes one of the more robust findings of our field and is one of the few phenomena that has undergone substantial empirical scrutiny. Moreover, it is one of the few phenomena that has been approached within a meta analytical analysis, offering a quantitative baseline to compare any future results.

3 Method

In the following, we describe the methodology for two studies that were aimed at replicating recent laboratory findings by Roettger et al. (2014, Experiment 1). Since the original data are available online (doi: 10.17605/osf.io/y7tdq), we can directly compare the estimated effect magnitudes as well as their precision. In §3.1 we briefly summarize the methodology employed in the original study and in §3.2 we discuss the context and methodology of our two classroom studies.

3.1 Roettger et al (2014)

In the following we briefly summarize relevant dimensions of the original study. For more detailed information about the method, see the original paper.

Sixteen native speakers of German participated in the experiment (corresponding to a mid to high-power regime according to Kirby & Sonderegger, 2018). All were students in the humanities living in the Cologne area. Most of them grew up in this area and all participants claimed to speak non-dialectal Standard German. The experiment was conducted in a sound-treated booth in a formal laboratory setting. On each trial, speakers heard a stimulus sentence such as (3) containing a nonce word inflected for plural and were subsequently asked to produce a corresponding sentence such as (4) containing a nonce word inflected for singular. Crucially, the singular form of the nonce contained a stop in word-final position and is thus subject to final devoicing.

(3) Plural stimulus
Aus Dortmund kamen die Drude.
'From Dortmund came the nonce-PL.'

(4) Singular response
Ein Drud wollte nicht mehr.
'One nonce-SG did not want to continue.'

The experimental items consisted of 24 pseudoword pairs, varying in terms of vowel quality and place of articulation of the critical stop. The duration of the vowels preceding the final stops were measured and submitted to statistical analysis. The authors report on a significant incomplete neutralization effect, i.e. vowels preceding devoiced stops were on average 9.1 ms longer than vowels preceding voiceless stops. The following two studies took Roettger et al's study as a point of departure and attempted to replicate their results.

3.2 Study 1 and 2

Study 1 and 2 were conducted by and for students during the summer terms 2015 and 2017 (henceforth Study 1 and 2). These studies were mainly intended as a pedagogical device to teach students about relevant and contemporary research question in phonetics and phonology, give them first hands-on experience in designing, conducting and analyzing experimental data, and to foster their first interpretation of empirical findings in light of the existing literature. The below described studies lack many aspects of control that we traditionally desire when conducting lab-based experiments (see §3.2.5 for a summary), which introduced a substantial amount of noise to the data. The increased level of noise should make it more difficult to detect a statistically robust signal in the data.

During the class, the lecturer (the second author) introduced the topic of final devoicing in German and its potential incomplete nature. The prevailing opposing views on incomplete neutralization were introduced, i.e. the assumption that incomplete neutralization is real and warrants explanation (e.g. Port & O'Dell, 1985) or that it is a methodological artifact (e.g. Fourakis & Iverson, 1984). As a response to the latter, methodological concerns were addressed (parallel to the discussion of Roettger et al. 2014). Most importantly, however, the recent experimental studies by Roettger et al. (2014) were not mentioned in class. Instead, a planned study was put forward as a potential resolution for previously mentioned methodological concerns. As far as we can tell, no student was aware of Roettger et al. (2014) until the ‘reveal’ at the end of the semester. In what followed, the lecturer directed students to design their own study, in which many of the above-mentioned problems would be avoided. Subsequently, students were guided to plan, design, run, and analyze an experiment that investigated incomplete neutralization.

The class spanned 12 weeks (one semester). In each week, there was one general lecture, giving theoretical and methodological background and a subsequent sub-seminar (10 and 11 different sub-seminars, respectively), in which students received practical training from more experienced student tutors (graduate students). Tutors guided the sub-seminar as part of their own teaching training or as student assistants.

3.2.1 Experimenter, participants, and procedure

In Study 1, 79 students recorded data from one speaker each. They were asked to run the experiment with a native German adult from their personal environment who is not involved in the class. The procedure resulted in 79 speakers (43 female, 36 male) with a mean age of 29 (ranging from 17 to 62). In Study 2, 65 students collected data from one speaker each, resulting in 65 participants (38 female, 27 male) with a mean age of 29 (ranging from 18 to 52). These numbers correspond to a sample size more than four times as large as the original study (n = 16), exceeding common recommendations for choice of sample size in replication studies (e.g. 2.5 times the original sample size: Simonsohn, 2015). Students, functioning as individual experimenters, were undergraduates majoring in general linguistics. They were in their second semester and had already passed an introduction lecture on phonetics. They had no background in experimental research.

3.2.2 Design

Following Roettger et al., students were guided to design a wug study (Berko, 1958) in which German speakers would be asked to derive singulars from given pseudoword plurals. Generally, speakers heard a stimulus sentence containing a nonce word inflected for plural and were subsequently asked to produce a corresponding sentence containing a nonce word inflected for singular. The singular form of the nonce contained a stop in word-final position.

Study 1 and 2 differed with respect to the context narrative that was spun around the task and presented to the speakers as part of the instructions. In Study 1, speakers were told that Noah was about to rescue some more animals which were still missing just before the departure of the arch. Speakers were instructed to listen to an audio playback telling them which animals were still missing (plural sentence: “Noah wollte schon mit seiner Arche ablegen. Es fehlten nur noch zwei Drude.” engl. ‘Noah wanted to set sail with the ark. Only two nonce-PL were missing.’) and to subsequently provide this information verbally (singular sentence: "Leider ist ein Drud ausgebüxt." engl. ‘Sadly, one nonce-SG ran away.’).

In Study 2, the story was about "Pakos", a new toy trend among little children. Speakers were instructed to listen to an audio playback in which a sales assistant tells the speaker where to find the specific toy (plural sentence: “In der Spielzeugabteilung gibt es eigene Aufsteller für den neuen Trend. Besonders beliebt sind die Drude.” engl. ‘In the toy department, there are stand-up displays for the new trend. Nonce-PL are particularly popular.’) and to subsequently provide this information verbally (singular sentence: "Ich habe einen Drud eingepackt." engl. ‘I took one nonce-SG.’).

The narrative was the same across all sub-seminars within each study, respectively. However, the stimuli within each sub-seminar differed. The narrative context alongside the instructions was presented in written form via powerpoint. After a familiarization phase with six real word examples (i.e. Kühe ‘cows’, Schafe ‘sheeps’, Hühner ‘chicken’, Elefanten ‘elephants’, Pinguine ‘penguins’, and Kamele ‘camels’), speakers went through the experimental trials in a self-paced manner. Stimuli were pseudorandomized.During the test phase, speakers’ productions were recorded via Audacity (Team Audacity, 2015), Praat (Boersma & Weenink, 2015) or other smartphone devices by individuals experimenters.

3.2.3 Materials

180 trochaic pseudowords of the shape $(C_1)C_2V_1C_3V_2$ served as stimuli. V_2 was always an unstressed schwa in order to mimic common German plural forms. All stimuli conformed to German phonotactics. In C_3position of target words, there was always a stop, balanced for place of articulation (labial, coronal, dorsal) and voicing status (voiced, voiceless). This resulted in 90 target words. The word onset $((C_1)C_2)$ was variable, with V_1 always being a tensed vowel (i.e. one of the following vowels (/i,a,u,o,e/).

Additionally, a list of 90 fillers was prepared by the lecturer. Fillers had the same phonotactic structure as targets. In fillers, C_3 was a sonorant (/l, m, n/) or a voiceless fricative (/f, ʃ/). Half of the fillers had a front vowel in V_1position (/y,ʏ,ø,œ,ɛ/) which can be interpreted as an umlaut. As plural forms with umlaut vowels sometimes do and sometimes do not require a vowel change (e.g., Türme > Turm ‘towers/tower’ but Hütten > Hütte ‘huts/hut’), we followed Roettger et al. (2014) and hoped their design would increase the salience of the fillers, simultaneously detracting attention from the critical stimuli.

All plural items were recorded in an anechoic booth at 48,000 Hz (Phantom 48V Microphone, Marantz Recorder pmd 570) by a native German speaker, a student of the sub-seminar, respectively.

3.2.4 Acoustic Analysis

The acoustic analyses were carried out by the students themselves. As part of the training, they were taught to identify and annotate segment boundaries for vowels using Praat (Boersma & Weenink, 2015/2017). The durations of the vowels preceding the final stops were measured by the first author. If the sound preceding the vowel/diphthong was a stop, the onset of the vowel was defined as the onset of voicing in cases of voiceless stop or as the end of the burst in cases of voiced stops. A sudden discontinuity in the spectrogram was taken as the onset of vowels following fricatives ([ʃ]), nasals ([m] and [n]), laterals ([l]) and palatal approximants ([j]) (e.g., [ʃuːk], [muːp], [bloːk] or [jiːt]). The end of the vowel was defined as the end of the second formant of the vowel, which usually coincided with a sudden drop in amplitude of voicing. The annotated data from each student were then distilled into one data table which was subsequently used for statistical analysis in the class.

3.2.5 Comparison to the original study

The experimental rationale and most of the design choices are comparable to Roettger et al.'s study. The general elicitation method (auditory plural-to-singular derivation), the structure and segmental make-up of the nonce words and the type of filler items are also comparable to the design choices of the original study. The instruction and the accompanying narrative, the actual nonce words and the speaker voices necessarily differ from the original study (as does, of course, the sample). These differences resemble common differences between original studies and direct replications. However, there are several differences between the original and the classroom studies that we would like to explicitly highlight.

Many factors controlled for in the original study were not controlled for in the classroom studies. Since each student recorded their own speaker, there were as many experimenters as there were experimental speakers. With each experimenter came a different recording environment, a different recording device (e.g. laptop or phone) and software (e.g. Audacity or Praat). Moreover, since certain steps of the experiment were assigned as homework, they were not cross-validated by the lecturer. Related to that, our experimenters/annotators had only little experience with conducting experiments, recording, and annotating speech data. All of these factors arguably introduce noise into our dataset. The increased level of noise makes it presumably more difficult to detect a signal in our data.

3.3 Statistical analysis

All data tables and R scripts are available here: https://osf.io/9kywf/. We submitted preceding vowel duration of all three experiments (Roettger et al.'s Experiment 1, Study 1 and Study 2) to Bayesian hierarchical models using the Stan modelling language (Carpenter et al., 2016) and the `R` package `brms` (Bürkner, 2016). We explicitly did not 'clean' the data according to any exclusion criteria. The potentially increased level of noise that is introduced by, for example, annotators with low proficiency should, in principle, make it more difficult to find any robust pattern in the data. We operate within the Bayesian inferential framework (rather than within a frequentist framework) for two reasons:

First, Bayesian methods allow us to directly answer the primary question: How plausible is our hypothesis given the data? We can answer this question by quantifying our uncertainty about the parameters of interest, which frees us from committing to hard cut-off points for statistical significance (such as the arbitrary 0.05 alpha level).

Second, it is easier to flexibly define hierarchical models (also known as mixed effects or multilevel models) in the Bayesian framework than in the frequentist framework. The frequentist linear mixed model standardly used in quantitative linguistics is generally fit with the `lme4` package (Bates et al., 2015b) in `R`. However, linear mixed effects models that also include the maximal random effect structure justified by the design (Barr et al., 2013; Schielzeth & Forstmeier, 2009) tend to not converge or to give unrealistic estimates of the correlations between random effects (Bates et al., 2015a). In contrast, the maximal random effects structure can be fit without problems using Bayesian hierarchical models (e.g. Bates et al., 2015a). Compared to non-maximal models using `lme4`, our maximal Bayesian models can be considered more conservative, making any detection of a difference due to spurious relationships less likely.

For Roettger et al.'s study, and both our Study 1 and 2, we fit hierarchical regression models to log-transformed vowel duration predicted by VOICING (voiceless vs. devoiced, dummy coded). The models included maximal random-effect structures, including a random intercept for target words and speakers (nested in sub-seminars for Study 1 and 2), and random slopes allowing the predictor to vary by target words and speakers (nested in sub-seminars for Study 1 and 2).

We used regularizing Gaussian priors (e.g. Gelman et al., 2008) centered around zero for all model parameters (i.e. our prior assumption is there is no effect of VOICING). Four sampling chains with 2000 iterations each were run for each model, with a warm-up period of 1000 iterations. We report, for each parameter of interest, 95% credible intervals and the posterior probability that a coefficient parameter is bigger than zero $Pr(\beta > 0)$. A 95% credible interval demarcates the range of values that comprise 95% of the probability mass of our posterior beliefs, such that no value inside the CI has a higher probability than any point outside of it (see, e.g., Jaynes & Kempthorne, 1976; Morey et al., 2016). We judge there to be compelling evidence for an effect if zero is (by a reasonably clear margin) not included in the 95% CI and $Pr(\beta > 0)$ is close to 0.

The posterior distributions for the effect of VOICING on the log scale were back-transformed to posterior distributions in milliseconds in order to compare to the meta analytical estimates by Nicenboim et al. (2018). Suppose that the fixed effects part of the model is defined as follows:

$$\log(rt) = \beta_0 + \beta_1 \text{VOICING} \quad (1)$$

This way we can obtain the posterior distribution of the difference in means between the two levels of VOICING by computing the difference between the intercept (β_0) and the intercept adjusted for the change in VOICING level: $\exp(\beta_0) - \exp(\beta_0 + \beta_1 \text{VOICING} \times 1)$.

As of yet, there is no single standard for evaluating replication success (Open Science Collaboration, 2014). Here we evaluated the replication of Roettger et al. using the posterior population estimates for the difference between devoiced and voiceless stops. We claim there to be a successful replication *of the general phenomenon of incomplete neutralization*, if the 95% of the probability mass of our posterior beliefs does not include zero and $Pr(\beta > 0)$ is close to 0. We claim there to be a successful replication *of the original study*, if the estimated means of our replication attempts fall within the 95% CI of the original estimate (Vasishth et al., 2018).

4 Results and discussion

4.1 Reanalysis of Roettger et al. (2014) as a baseline

Figure 1 shows the observed differences between devoiced and voiceless stops in Roettger et al.'s Experiment 1. The semitransparent dots are individual aggregates. As the figure clearly indicates, the probability mass for both speakers and items is above zero (dashed line) with most speakers/items showing an incomplete neutralization effect.

Our Bayesian inferential assessment speaks to these descriptive observations: Based on the data, there is substantial evidence for vowels preceding devoiced stops having greater duration with an effect size of 0.22 standard deviations ($\beta = 9.1$, 95%CI = [4.3,13.8]; $Pr(\beta<0)$ = 0.0005). As expected, we were able to reproduce Roettger et al. (2014) original analysis, leading us to conclude that speakers in this study produced larger durations for vowels preceding devoiced stops. These standardized values can now function as a baseline against which we can compare the results of Study 1 and 2. We first discuss the individual analyses for both Study 1 and 2, individually and then compare them to the original study.

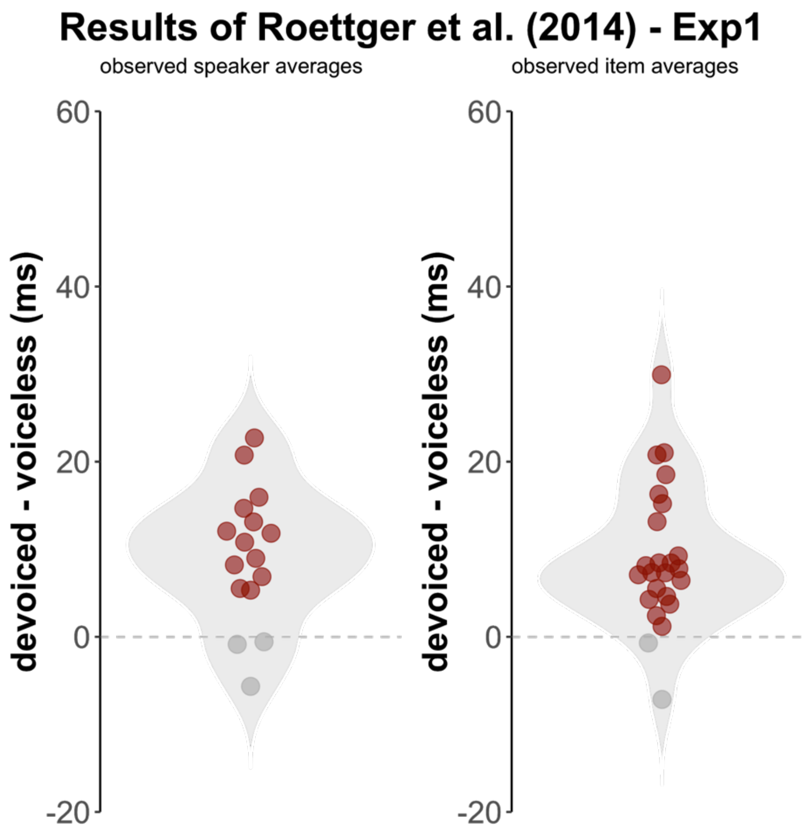

Figure 1: Observed difference between devoiced and voiceless stops in Roettger et al.'s Experiment 1 for individual speakers (left panel) and individual items (right panel). Semitransparent red dots indicate values that are compatible with incomplete neutralization effects, i.e. vowels preceding devoiced stops exhibit greater duration than vowels preceding voiceless stops. The gray density shape in the background represents the probability density along the measurement.

4.2 Results of Experiment 1

Figure 2 shows the observed differences between devoiced and voiceless stops in Study 1. Despite a substantial number of data points indicating a reversed effect (see also Roettger et al. 2014), there is clearly a stochastic dominance pointing towards incomplete neutralization (i.e. red dots above zero) for the sample of both speakers and target words.

Our Bayesian inferential assessment speaks to these descriptive observations: Based on the data, there is substantial evidence for vowels preceding devoiced stops having greater durations (β = 6.4, 95% CI [3.6, 9.3], $Pr(\beta<0)$ = 0.0003). We can thus claim to have replicated an incomplete neutralization effect.

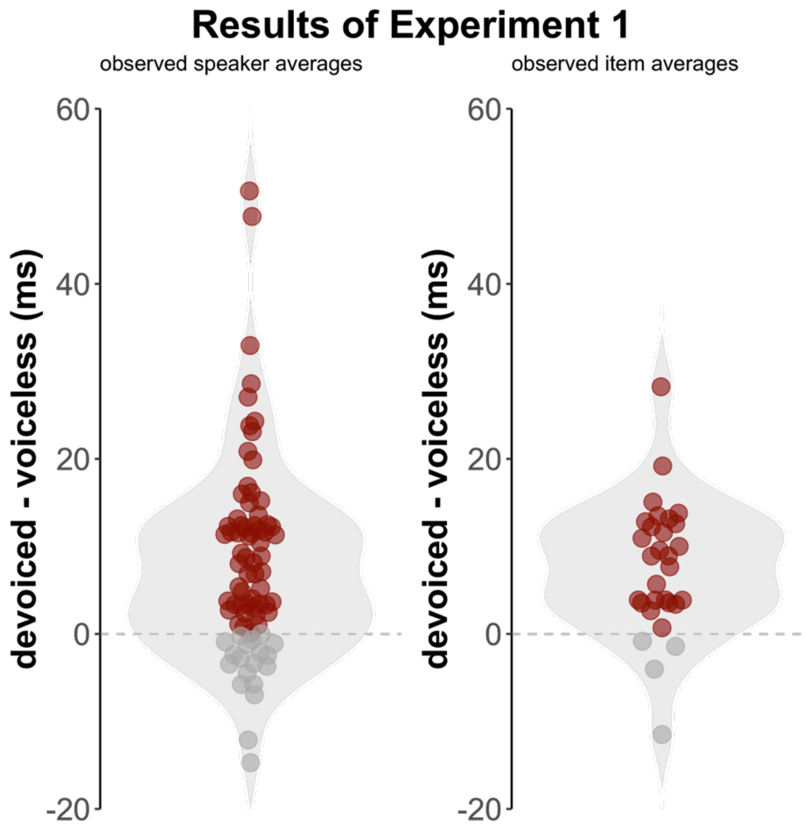

Figure 2: Observed difference between devoiced and voiceless stops in Study 1 for individual speakers (left panel) and individual items (right panel). Semitransparent red dots indicate values that are compatible with incomplete neutralization effects, i.e. vowels preceding devoiced stops exhibit greater duration than vowels preceding voiceless stops. The gray density shape in the background represents the probability density along the measurement.

4.3 Results of Experiment 2

Figure 3 shows the observed differences between devoiced and voiceless stops in Study 2. As before, the figure clearly indicates that the probability mass for both speakers and items is above zero (dashed line) with most speakers/items showing an incomplete neutralization effect.

Again, our Bayesian inferential assessment confirms these descriptive observations: Based on the data, there is substantial evidence for vowels preceding devoiced stops having greater duration (β = 9.3, 95% CI = [5.6, 13], $Pr(\beta < 0) \approx 0$). Experiment 2 has replicated an incomplete neutralization effect. Now this raises the question as to how well these two studies have replicated the original study by Roettger et al. (2014)?

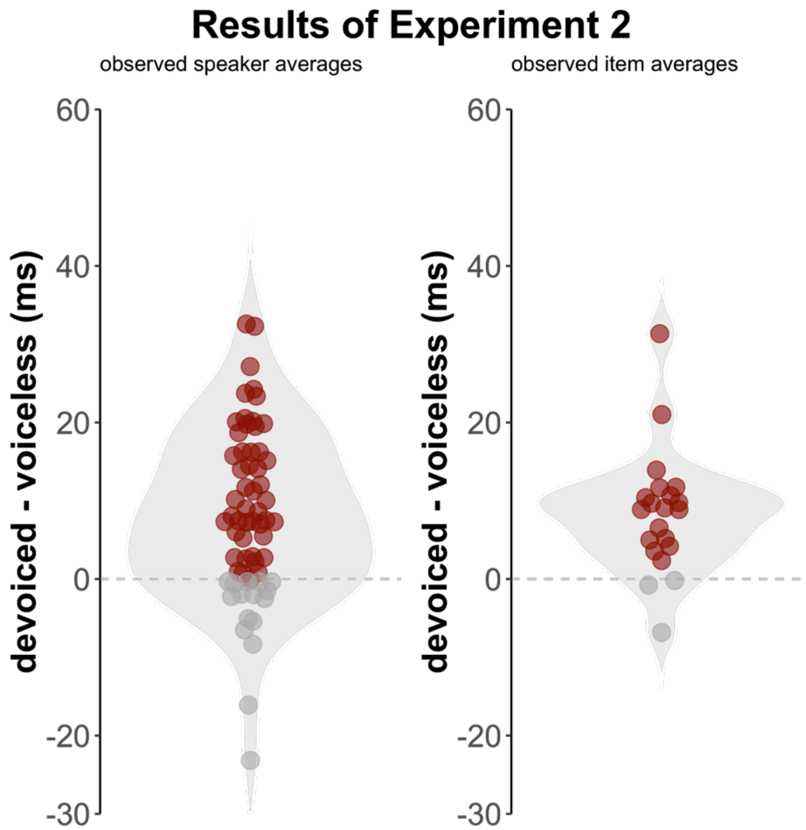

Figure 3: Observed difference between devoiced and voiceless stops in Study 2 for individual speakers (left panel) and individual items (right panel). Semitransparent red dots indicate values that are compatible with incomplete neutralization effects, i.e. vowels preceding devoiced stops exhibit greater duration than vowels preceding voiceless stops. The gray density shape in the background represents the probability density along the measurement.

4.4 Comparison to Roettger et al. (2014)

While we have replicated incomplete neutralization of final devoicing with similar numerical differences between devoiced and voiceless stops to other lab-based studies including Roettger et al. (2014) and meta analytical estimates by Nicenboim et al. (2018), the estimated effect magnitude is very small. This raises the question as to how well our new findings replicate Roettger et al.'s findings? The Bayesian framework allows us to quantify the answer to this question by comparing the posterior distributions. As specified above, we claim there to be a successful replication of the original study, if the estimated means of our replication attempts fall within the 95% CI of the original estimates (Vasishth et al., 2018).

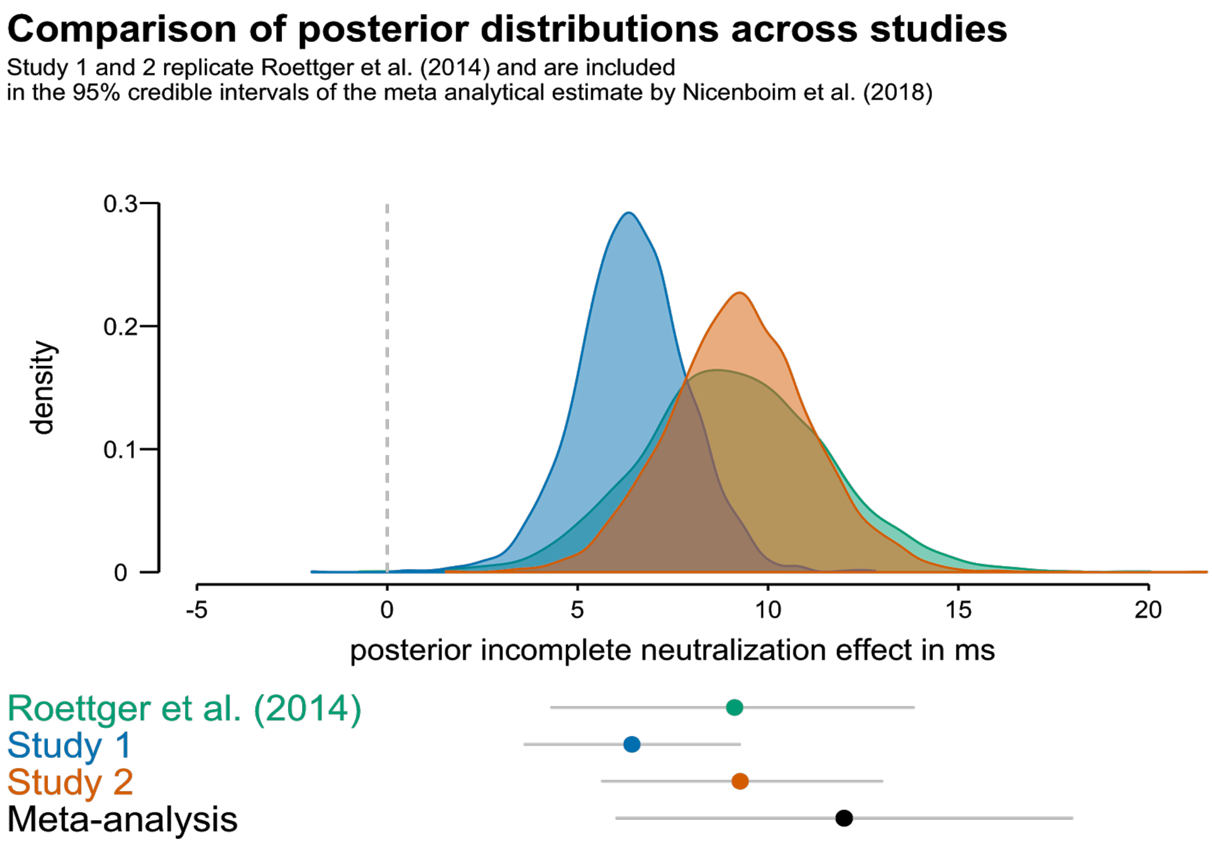

Figure 4: Posterior distributions of the incomplete neutralization effect (duration of vowel preceding devoiced stop - duration of vowel preceding voiceless stop) for all three data sets. The points below the density curves indicate the posterior means and 95% credible intervals for all three studies and the meta analytical estimate from Nicenboim et al. (2018: 50).

Fig. 4 illustrates the high level of overlap between all three studies. Study 1 overall suggests a smaller magnitude of the incomplete neutralization effect with higher precision (i.e. uncertainty about our estimate of interest) than both Study 2 and Roettger et al.'s original study. The posterior distribution of Study 2 almost entirely overlaps with the posterior distribution of Roettger et al.'s study, exhibiting a similar location with slightly higher precision. As the point and whiskers below the distributions in Fig. 4 suggest, both the mean of Study 1 and 2 fall within the 95% CI of Roettger et al.'s data, as well as the 95% CI of the meta-analysis presented in Nicenboim et al. (2018).

We can thus conclude that our replications in the classroom were successful. Despite a low level of control over factors commonly controlled for in the laboratory, the classroom replications show an even higher precision than their laboratory counterpart. This increased precision is mainly achieved by a larger sample (16 speakers in Roettger et al. vs. 79/65 speakers in Study 1 and 2). Given the more feasible data collection procedure, such large sample sizes can be easily achieved within the classroom scenario.

5 General discussion

5.1 Summary

The aforementioned studies replicated incomplete neutralization of final devoicing in German. The vowels preceding so-called ‘devoiced’ stops, i.e. stops derived from a morphological paradigm that contains voiced stops intervocalically, exhibit a greater duration than voiceless stops. The design of our studies closely resembled the experimental paradigm used by Roettger et al. (2014), a high-powered laboratory-based incomplete neutralization study in German.

Beyond simply replicating incomplete neutralization (i.e. merely testing whether a difference of 0 is plausible given the model and the data), we were able to replicate the original magnitude of the effect by Roettger et al. (2014). It is important to replicate and therefore substantiate the magnitude of an acoustic effect, because there are limits to what acoustic difference listeners can detect reliably. For durational differences for example, it is commonly assumed that listeners can reliable hear difference of 10 ms upwards (e.g., Hirsh, 1959; Lehiste, 1976; Phillips et al., 1994; Klatt, 1976; Miller, 1989). As the statistical analyses of Study 1, 2, and Roettger et al. above have shown, values above 10 ms are plausible given the data and the model, however values below 10 ms are even more plausible. Nicenboim et al.’s (2018) meta-analysis suggest likely values between 6 and 18 ms, but, as the authors discuss in detail, these values might be skewed towards upper values due to publication bias (Sterling, 1959) and Type-M errors (i.e. an overestimation of effect magnitudes due to small sample sizes, Gelman & Carlin, 2014). While, we can be more and more certain that there is a difference between neutralized and non-neutralized sounds, the jury is still out on whether the magnitude of the difference is likely to play any role in speech communication (Kohler, 2012; Winter & Roettger, 2011; Roettger et al., 2014). If these differences are relevant for aspects of communication, we should consider them within our models of phonological knowledge representations (e.g. Dinnsen & Charles-Luce, 1984; Charles-Luce, 1985; Port & O’Dell, 1985; Oostendorp, 2008). If they are not, incomplete neutralization might be better interpreted as an artifact of lexical organization and spreading activation (e.g., Ernestus & Baayen, 2006; Goldrick, Folk, & Rapp, 2010; Kleber et al., 2010; Winter & Roettger, 2011; Roettger et al., 2014; Seyfarth et al., 2018).

Regardless of its interpretation, the present studies, taken together with the published literature on this topic, invite us to be more and more certain about incomplete neutralization. Given the relatively large literature, its recent meta-analytical assessment and the presence of numerous replication attempts, including our own, incomplete neutralization of final devoicing has become one of the most robust phenomena within experimental phonetics.

5.2 Creating a replication landscape

For the present paper, incomplete neutralization was chosen for illustrative purposes. By way of example, we argue that direct replications of published experiments do not need to be resource-intensive, but can be done in the classroom in collaboration with our students. This concept makes replications feasible in terms of overall cost/benefit ratios. Ultimately, this approach will hopefully lead to more replication studies being conducted. This is important because the recent crisis in confidence within the social sciences and their associated replication failures demonstrated how misleading published results can be. While the awareness of these issues has caused an ongoing and intensive discourse across academic disciplines and has already led to methodological advancements and increased rigor in trying to ensure reliability of scientific findings, there is still much work to be done. One important step to ensure scientific reliability is to conduct direct replication studies and to create an environment in which such replication studies are incentivized.

Although resource-intensive multi-site replication attempts will remain the gold standard (Open Science Collaboration, 2015; Frank et al., 2017; Nieuwland et al., 2018), more and more people have started to suggest more accessible ways to conduct replication studies (Frank & Saxe, 2012; Grahe et al., 2014, 2016; Hawkins et al., 2018).

As we have also argued in this paper involving our students can be a valid way forward for speech production research and can be considered a win-win for both our students and our research community. This approach has several noteworthy educational benefits: The fact that the original study was published suggests that the authors asked a relevant research question. Spending time and effort into replicating their results is thus plausible and justified and gives the students an opportunity to make a real contribution to the field. While the hardest part of research projects is often finding a research question and operationalizing a way to put this question to the test, taking the original study as a departure point sidesteps the important conceptual work, allowing students to focus solely on the methodology. After having run the experiment and analyses, students will be rewarded with on of two important experiences: Either they are able to replicate the original results or they are not. In either case, they will wonder about the outcome and will reflect upon important aspects of the scientific machinery.

In addition to these pedagogical advantages, the research community would get a large number of free workers. Students sit in our method classes anyways, we might as well utilize them for a greater scientific good.

Beyond finding affordable ways of conducting replication studies as suggested here, we also need to reconsider our incentive systems. The problem is that there are currently few mechanisms in place to encourage replication attempts. For example, Martin and Clarke (2017) report survey results, suggesting that, in 2015, only 3% of psychology journals explicitly state that they will consider publishing replications.

To overcome the asymmetry between the cost of direct replication studies and the presently low academic payoff for it, we as a research community must re-evaluate the value of direct replications. Funding agencies, journals, editors, and reviewers should start valuing direct replication attempts, be it successful replications or replication failures, as much as they value novel findings. For example, we could either dedicate existing journal space to direct replications (e.g. as an article type) or by creating new journals that are specifically dedicated to replication studies. Complementary, we could implement Replication Awards by communities, institutes or journals (Gorgolewski et al., 2018) or encourage Ph.D. students to replicate existing findings as their first step towards approaching any empirical finding ('replication-first rule', Kochari & Ostarek, 2018)

As soon as we make publishing replications easier, more researchers will be compelled to replicate both their own work and the work of others. Only by replicating empirical results and evaluating the accumulated evidence can we substantiate previous findings and extend their external validity. Thus, the present proposal is only one of many strategies to counter the upcoming crisis in confidence and to offer a solution to the cost-reward asymmetry that is currently in place for replication studies. Our proposed strategy, replications in the classroom, however come with their own set of limitations and challenges which we will address in the following.

5.3 Limits and challenges of this approach

While there are ongoing debates surrounding the questions of experimental control vs. stylistic variation (Xu, 2010; Wagner et al., 2015 and references therein), an increasing number of speech scientist started to use crowdsourcing platforms to collect data (Hasegawa-Johnson et al., 2015). This suggests to us that researchers are willing to trade an increased level of variability and lack of control with quick and convenient access to large samples. One concern that arises surrounds the question as to whether the data collected in the classroom is of sufficient quality?

Our students were neither experienced experimenters nor trained annotators. Their limited domain expertise might be cause of concern as to how valid the annotated acoustic intervals are. The amount of variability within and across annotators clearly exceeded the more homogenous patterns found in laboratory studies with experienced annotators. But within these limitations, we were able to produce two relatively close replication studies with sufficient precision. Again, serving as an existence proof, our studies demonstrate what is possible for motivated students using freely available tools and resources. In fact, students might be very careful and motivated, fostering surprisingly precise analyses. Based on feedback from the students, it seems highly motivating for students to know that they were running experiments not purely as an exercise that is eventually marked but for the purpose of evaluating existing findings that have been presented by professional researchers and that have been published in scientific journals. If we can validate the students' work this way, this might have a positive effect on their diligence and motivation. We acknowledge that the quality of replications in the classroom will heavily depend on the students' motivation, effort and care as well as on their supervision by the instructors. In light of these degrees of freedom, the instructor ultimately needs to evaluate the quality of the research and if found suitable, should be the leading communicator of the results. Hawkins et al. (2018) describe that over half of the projects conducted in their courses met their quality criterion. Keep in mind that without any post hoc quality control in the present data set, we still replicated results closely compatible with both controlled laboratory experiments and recent meta-analytical estimates. It is also noteworthy that any of these concerns related to the research quality can be made for published research in general.

Beyond quality control, there are important limitations of the proposed classroom replications which are particularly relevant for speech related research. The experiments in the classroom must stay within the technical abilities of the students, the practical limits of a university class, and the limits of accessible populations. There are many experimental procedures that warrant substantially more expertise to execute than speech recording within a simple powerpoint presentation. These limitations also often come with their own restrictions on technical abilities and equipment. While more complex experimental set ups such as eye-tracking and mouse tracking experiments might still be feasible to some extent, neurolinguistic experiments (e.g. using electroencephalography or functional magnetic resonance imaging) and articulatory studies (e.g. using ultrasound or electromagnetic articulography), are probably often beyond the realm of classroom replications. These techniques not only require access to expensive lab space and equipment, but also require substantial training and technical know-how. Additionally, student replications will be bounded by the temporal structure of a semester and the populations that are accessible to the students. This translates into another area of quantitative linguistics that will not be within the limits of classroom replications: Most field methods that require access to a particular speaker population and/or geographical locations will often not be feasible targets to replicate. We thus need to acknowledge that our proposed strategy will inevitably be biased towards a subset of behavioral research. In summary, replications in the classroom will be restricted to easy studies that do not require complex experimental set ups, can be executed within a short period of time and be conducted on an accessible population.

Again, it is noteworthy that these biases are probably present in all our published research, as the trade-off between resources and academic reward are important driving forces for the majority of researchers. Moreover, many of our theories are founded on rather straightforward speech production and perception experiments, most of which may qualify for exactly these types of classroom experiments. We want to emphasize again that, while the proposed method is not a single panacea for the present crisis of confidence, we firmly believe it is one possible step towards a more substantiated scientific record within experimental linguistics.

5.4 Where to go from here

The presented speech production studies were replications of published experiments. Although their protocols were following the original design very closely, they also differed in certain ways. While it can be argued that differences in the material lead to an ecologically more valid extension of the original findings, these differences can also be used to doubt the meaningfulness of the results in relation to the original study. In the future, it would be desirable to stick even closer to the original study. One could contact the original author and ask for their materials as well as requesting further insights about the procedure that go beyond the reported details in the manuscript.

We could further improve our protocol by preregistering our replication attempt. A preregistration is a time-stamped document in which researchers specify exactly how they plan to collect their data and how they plan to conduct their confirmatory analyses. Preregistrations can be a powerful tool to reduce exploitation of researcher degrees of freedom (Simmons et al., 2011; Roettger, 2018) because researchers are required to commit to certain decisions prior to observing the data. Additionally, public preregistration can at least help to reduce issues related to publication bias, i.e. the tendency to publish positive results more often than null results (Franco et al., 2014; Sterling, 1959), as the number of failed attempts to reject a hypothesis can be tracked transparently.

There are several websites that offer services and / or incentives to preregister studies prior to data collection, such as AsPredicted (AsPredicted.org) and the Open Science Framework (osf.io). These platforms allow us to time-log reports and either make them publicly available or grant anonymous access only to a specific group of people (such as reviewers and editors during the peer-review process).

Preregistering our replication attempts (in the classroom) could further help improving the quality of our methodology as the instructor can evaluate the written preregistration protocol prior to data collection and thus suggest changes to the proposed method and review analytic code for errors.

6 Conclusion

The presented results demonstrate the practical possibility of performing replication research on speech phenomena in the classroom. Although associated with a certain lack of control and an increased level of noise within the data, replicating in the classroom offers an easy and cheap way to replicate relevant experimental findings of our literature. Beyond the possibility of substantiating our scientific record, this technique has numerous pedagogical advantages for our students.

Still, there are many challenges we need to pay close attention to. We need to find ways to ensure high-quality data. Close supervision of our students by an experienced instructor and preregistration of the methodological protocols can help ensure this quality. Moreover, due to practical considerations, replications in the classroom remain biased towards a subset of phonetic research, restricted to simple studies on accessible population. Despite these challenges and limitations, we believe that if this approach were implemented more widely across our field, the scientific and educational pay-off would be substantial.

References

Allaire, J. J., Wickham, H., Ushey, K., & Ritchie, G. (2017). rstudioapi: Safely Access the RStudio API (Version 0.7). Retrieved from https://CRAN.R-project.org/package=rstudioapi

Audacity Team. (2015). Audacity (R): Free audio editor and recorder [computer application]. (Version 2.1.0). Retrieved from https://audacityteam.org/

Auguie, B. (2017). gridExtra: Miscellaneous Functions for “Grid” Graphics (Version 2.3). Retrieved from https://CRAN.R-project.org/package=gridExtra

Barr, D. J., Levy, R., Scheepers, C., & Tily, H. J. (2013). Random effects structure for confirmatory hypothesis testing: Keep it maximal. *Journal of Memory and Language*, *68*(3), 255–278.

Bates, D., Kliegl, R., Vasishth, S., & Baayen, H. (2015). Parsimonious mixed models. *Eprint ArXiv:1506.04967*.

Bates, D., Mächler, M., Bolker, B., & Walker, S. (2015). Fitting Linear Mixed-Effects Models Using lme4. *Journal of Statistical Software*, *67*(1), 1–48. https://doi.org/10.18637/jss.v067.i01

Behrend, T. S., Sharek, D. J., Meade, A. W., & Wiebe, E. N. (2011). The viability of crowdsourcing for survey research. *Behavior Research Methods*, *43*(3), 800–813. https://doi.org/10.3758/s13428-011-0081-0

Berko, J. (1958). The child’s learning of English morphology. *Word*, *14*(2–3), 150–177.

Boersma, P., & Weenink, D. (2015). Praat: doing phonetics by computer [Computer program] (Version Version 5.4.12, 2015).

Bürkner, P.-C. (2017). brms: An R package for Bayesian multilevel models using Stan. *Journal of Statistical Software*, *80*(1), 1–28. https://doi.org/10.18637/jss.v080.i01

Camerer, C. F., Dreber, A., Forsell, E., Ho, T.-H., Huber, J., Johannesson, M., … Wu, H. (2016). Evaluating replicability of laboratory experiments in economics. *Science*, *351*(6280), 1433–1436. https://doi.org/10.1126/science.aaf0918

Campbell, D. T. (1969). Reforms as experiments. *American Psychologist*, *24*(4), 409.

Carney, D. R., Cuddy, A. J. C., & Yap, A. J. (2010). Power posing: Brief nonverbal displays affect neuroendocrine levels and risk tolerance. *Psychological Science*, *21*(10), 1363–1368.

Carpenter, B., Gelman, A., Hoffman, M. D., Lee, D., Goodrich, B., Betancourt, M., … Riddell, A. (2017). Stan: A probabilistic programming language. *Journal of Statistical Software*, *76*(1).

Charles-Luce, J. (1985). Word-final devoicing in German and the effects of phonetic and sentential contexts. *The Journal of the Acoustical Society of America*, *77*.

Clarke, E., & Sherrill-Mix, S. (2017). Ggbeeswarm: Categorical scatter (violin point) plots (Version 0.6.0). Retrieved from https://CRAN.R-project.org/package=ggbeeswarm

DeLong, K. A., Urbach, T. P., & Kutas, M. (2005). Probabilistic word pre-activation during language comprehension inferred from electrical brain activity. *Nature Neuroscience*, *8*(8), 1117–1121.

Dinnsen, D. A., & Charles-Luce, J. (1984). Phonological neutralization, phonetic implementation and individual differences. *Journal of Phonetics*, *12*(1), 49–60.

Dinnsen, D., & Garcia-Zamor, M. (1971). The three degrees of vowel length in German. *Paper in Linguistics*, *4*(1), 111–126. https://doi.org/10.1080/08351817109370250

Doyen, S., Klein, O., Pichon, C.-L., & Cleeremans, A. (2012). Behavioral priming: It's all in the mind, but whose mind? *PLOS ONE*, *7*(1), e29081. https://doi.org/10.1371/journal.pone.0029081

Dunlap, K. (1925). The experimental methods of psychology. *The Pedagogical Seminary and Journal of Genetic Psychology*, *32*(3), 502–522.

Ernestus, M., & Baayen, R. H. (2006). The functionality of incomplete neutralization in Dutch: The case of past-tense formation. *Laboratory Phonology*, *8*(1), 27–49.

Errington, T. M., Iorns, E., Gunn, W., Tan, F. E., Lomax, J., & Nosek, B. A. (2014). Science forum: An open investigation of the reproducibility of cancer biology research. *ELife*, *3*, e04333.

Fine, A. B., Jaeger, T. F., Farmer, T. A., & Qian, T. (2013). Rapid expectation adaptation during syntactic comprehension. *PLOS ONE*, *8*(10), e77661.

Fourakis, M., & Iverson, G. K. (1984). On the 'incomplete neutralization' of German final obstruents. *Phonetica*, *41*(3), 140–149.

Franco, A., Malhotra, N., & Simonovits, G. (2014). Publication bias in the social sciences: Unlocking the file drawer. *Science*, *345*(6203), 1502–1505.

Frank, M. C., Bergelson, E., Bergmann, C., Cristia, A., Floccia, C., Gervain, J., … Yurovsky, D. (2017). A collaborative approach to infant research: Promoting reproducibility, best practices, and theory-building. *Infancy*, *22*(4), 421–435. https://doi.org/10.1111/infa.12182

Frank, M. C., & Saxe, R. (2012). Teaching replication. *Perspectives on Psychological Science*, *7*(6), 600–604.

Fuchs, S. (2005). *Articulatory correlates of the voicing contrast in alveolar obstruent production in German*. Berlin: Zentrum für Allgemeine Sprachwissenschaften.

Gelman, A., & Carlin, J. (2014). Beyond power calculations assessing type S (sign) and type M (magnitude) errors. *Perspectives on Psychological Science*, *9*(6), 641–651. https://doi.org/10.1177/1745691614551642

Gelman, A., Jakulin, A., Pittau, M. G., Su, Y.-S., & others. (2008). A weakly informative default prior distribution for logistic and other regression models. *The Annals of Applied Statistics*, *2*(4), 1360–1383. https://doi.org/10.1214/08-AOAS191

Gelman, A., & Loken, E. (2014). The statistical crisis in science. *American Scientist*, *102*(6), 460–465.

Goldrick, M., Folk, J. R., & Rapp, B. (2010). Mrs. Malaprop's neighborhood: Using word errors to reveal neighborhood structure. *Journal of Memory and Language*, *62*(2), 113–134. https://doi.org/10.1016/j.jml.2009.11.008

Goodman, J. K., Cryder, C. E., & Cheema, A. (2012). Data collection in a flat world: The strengths and weaknesses of Mechanical Turk samples. *Journal of Behavioral Decision Making*, *26*(3), 213–224. https://doi.org/10.1002/bdm.1753

Gorgolewski, K. J., Nichols, T., Kennedy, D. N., Poline, J.-B., & Poldrack, R. A. (2018). Making replication prestigious. *Behavioral and Brain Sciences*, *41*, E131. https://doi.org/10.1017/S0140525X18000663

Grahe, J., Brandt, M., IJzerman, H., & Cohoon, J. (2014). Replication education. *APS Observer*, *27*(3).

Grahe, J., Brandt, M. J., IJzerman, H., Cohoon, J., Peng, C., Detweiler-Bedell, B., & Weisberg, Y. (2016). Collaborative replications and education project (CREP). *Retrieved from Osf. Io/Wfc6u*. https://doi.org/10.17605/OSF.IO/WFC6U

Grahe, J., Brandt, M., Wagge, J., Legate, N., Wiggins, B., Christopherson, C., … LePine, S. (2018). Collaborative replications and education project (CREP). https://doi.org/10.17605/osf.io/wfc6u

Hasegawa-Johnson, M., Cole, J., Jyothi, P., & Varshney, L. R. (2015). Models of dataset size, question design, and cross-language speech perception for speech crowdsourcing

applications. *Laboratory Phonology*, *6*(3–4), 381–431. https://doi.org/10.1515/lp-2015-0012
Hawkins, R. X., Smith, E. N., Au, C., Arias, J. M., Catapano, R., Hermann, E., … Reynolds, J. (2018). Improving the Replicability of Psychological Science Through Pedagogy. *Advances in Methods and Practices in Psychological Science*, *1*(1), 7–18. https://doi.org/10.1177/2515245917740427
Hewitt, J. K. (2012). Editorial policy on candidate gene association and candidate gene-by-environment interaction studies of complex traits. *Behavior Genetics*, *42*(1), 1–2. https://doi.org/10.1007/s10519-011-9504-z
Hirsh, I. J. (1959). Auditory perception of temporal order. *The Journal of the Acoustical Society of America*, *31*(6), 759–767. https://doi.org/10.1121/1.1907782
Inozuka, E. (1991). The realization of the German neutralized word-final plosives /g, k/ : An acoustic analysis. *Sophia Linguistica*, *30*, 119–134.
Ioannidis, J. P. (2005). Why most published research findings are false. *PLoS Medicine*, *2*(8), e124.
Jaynes, E. T., & Kempthorne, O. (1976). Confidence intervals vs. Bayesian Intervals. In W. L. Harper & C. A. Hooker (Eds.), *Foundations of Probability Theory, Statistical Inference, and Statistical Theories of Science*(Vol. 6b, pp. 175–257). Dordrecht: Springer.
Jespersen, O. (1920). *Lehrbuch der Phonetik: Mit 2 Tafeln*. (H. Davidsen, Trans.). Leipzig: BG Teubner.
Jessen, M., & Ringen, C. (2002). Laryngeal features in German. *Phonology*, *19*(2), 189–218. https://doi.org/10.1017/S0952675702004311
Kirby, J., & Sonderegger, M. (2018). Mixed-effects design analysis for experimental phonetics. *Journal of Phonetics*, *70*, 70–85. https://doi.org/10.1016/j.wocn.2018.05.005
Klatt, D. H. (1976). Linguistic uses of segmental duration in English: Acoustic and perceptual evidence. *The Journal of the Acoustical Society of America*, *59*(5), 1208–1221. https://doi.org/10.1121/1.380986
Kochari, A. R., & Ostarek, M. (2018). Introducing a replication-first rule for Ph.D. projects. *Behavioral and Brain Sciences*, *41*, E138. https://doi.org/10.1017/S0140525X18000730
Kohler, K. J. (2007). Beyond laboratory phonology - The phonetics of speech communication. In M.-J. Solé, P. S. Beddor, & M. Ohala (Eds.), *Experimental approaches to phonology*(pp. 41–53). New York: Oxford University Press.
Kohler, K. J. (2012). Neutralization?! The phonetics–phonology issue in the analysis of word–final obstruent voicing. In D. Gybbon, D. Hirst, & N. Campbell (Eds.), *Rhythm, melody and harmony in speech. Studies in honour of Wiktor Jassem.*(pp. 171–180). Poznań: Polskie Towarzystwo Fonetyczne.
Koole, S. L., & Lakens, D. (2012). Rewarding replications: A sure and simple way to improve psychological science. *Perspectives on Psychological Science*, *7*(6), 608–614.
Ladefoged, P. (2003). *Phonetic data analysis: An introduction to fieldwork and instrumental techniques*. Malden, MA: Blackwell.
Lehiste, I. (1976). Suprasegmental features of speech. In N. J. Lass (Ed.), *Contemporary issues in experimental phonetics*(Vol. 225, pp. 225–239). New York: Academic Press.
Lisker, L. (1986). ‘Voicing’ in English: A catalogue of acoustic features signaling /b/ versus /p/ in trochees. *Language and Speech*, *29*(1), 3–11. https://doi.org/10.1177/002383098602900102
Madden, C. S., Easley, R. W., & Dunn, M. G. (1995). How journal editors view replication research. *Journal of Advertising*, *24*(4), 77–87. https://doi.org/10.1080/00913367.1995.10673490
Makel, M. C., Plucker, J. A., & Hegarty, B. (2012). Replications in psychology research: How often do they really occur? *Perspectives on Psychological Science*, *7*(6), 537–542.

Manaster Ramer, A. (1996). A letter from an incompletely neutral phonologist. *Journal of Phonetics*, *24*(4), 477–489. https://doi.org/10.1006/jpho.1996.0026
Martin, G. N., & Clarke, R. M. (2017). Are psychology journals anti-replication? A snapshot of editorial practices. *Frontiers in Psychology*, *8*, 523. https://doi.org/10.3389/fpsyg.2017.00523
Mason, W., & Suri, S. (2012). Conducting behavioral research on Amazon's Mechanical Turk. *Behavior Research Methods*, *44*(1), 1–23. https://doi.org/10.3758/s13428-011-0124-6
Miller, J. D. (1989). Auditory-perceptual interpretation of the vowel. *The Journal of the Acoustical Society of America*, *85*(5), 2114–2134. https://doi.org/10.1121/1.397862
Mitleb, F. M. (1981). Temporal correlates of "voicing" and its neutralization in German. *Research in Phonetics*, *2*, 173–191.
Moreno, S. G., Sutton, A. J., Ades, A., Stanley, T. D., Abrams, K. R., Peters, J. L., & Cooper, N. J. (2009). Assessment of regression-based methods to adjust for publication bias through a comprehensive simulation study. *BMC Medical Research Methodology*, *9*(1), 2. https://doi.org/10.1186/1471-2288-9-2
Nicenboim, B., Roettger, T. B., & Vasishth, S. (2018). Using meta-analysis for evidence synthesis: The case of incomplete neutralization in German. *Journal of Phonetics*, *70*, 39–55.
Nicenboim, B., & Vasishth, S. (2016). Statistical methods for linguistic research: Foundational ideas - Part II. *Language and Linguistics Compass*, *10*(11), 591–613. https://doi.org/10.1111/lnc3.12207
Nieuwland, M. S., Politzer-Ahles, S., Heyselaar, E., Segaert, K., Darley, E., Kazanina, N., … Huettig, F. (2018). Large-scale replication study reveals a limit on probabilistic prediction in language comprehension. *ELife*, *7*, e33468. https://doi.org/10.7554/eLife.33468.001
Nosek, B. A., Spies, J. R., & Motyl, M. (2012). Scientific utopia: II. Restructuring incentives and practices to promote truth over publishability. *Perspectives on Psychological Science*, *7*(6), 615–631.
Open Science Collaboration. (2015). Estimating the reproducibility of psychological science. *Science*, *349*(6251). https://doi.org/10.1126/science.aac4716
Phillips, S. L., Gordon-Salant, S., Fitzgibbons, P. J., & Yeni-Komshian, G. H. (1994). Auditory duration discrimination in young and elderly listeners with normal hearing. *Journal of the American Academy of Audiology*, *5*, 210–215.
Piroth, H. G., & Janker, P. M. (2004). Speaker-dependent differences in voicing and devoicing of German obstruents. *Journal of Phonetics*, *32*(1), 81–109. https://doi.org/10.1016/S0095-4470(03)00008-1
Port, R. F., & Crawford, P. (1989). Incomplete neutralization and pragmatics in German. *Journal of Phonetics*, *17*, 257–282.
Port, R. F., & O'Dell, M. L. (1985). Neutralization of syllable-final voicing in German. *Journal of Phonetics*, *13*(4), 455–471.
Ranehill, E., Dreber, A., Johannesson, M., Leiberg, S., Sul, S., & Weber, R. A. (2015). Assessing the robustness of power posing: No effect on hormones and risk tolerance in a large sample of men and women. *Psychological Science*, *26*(5), 653–656. https://doi.org/10.1177/0956797614553946
Roettger, T. B. (2018). Researcher degrees of freedom in phonetic sciences. https://doi.org/10.31234/osf.io/fp4jr
Roettger, T. B., Winter, B., Grawunder, S., Kirby, J., & Grice, M. (2014). Assessing incomplete neutralization of final devoicing in German. *Journal of Phonetics*, *43*, 11–25. https://doi.org/10.1016/j.wocn.2014.01.002
Roettger, T., & Gordon, M. (2017). Methodological issues in the study of word stress correlates. *Linguistics Vanguard*, *3*(1). https://doi.org/10.1515/lingvan-2017-0006

Schielzeth, H., & Forstmeier, W. (2008). Conclusions beyond support: overconfident estimates in mixed models. *Behavioral Ecology*, *20*(2), 416–420. https://doi.org/10.1093/beheco/arn145

Seyfarth, S., Garellek, M., Gillingham, G., Ackerman, F., & Malouf, R. (2018). Acoustic differences in morphologically-distinct homophones. *Language, Cognition and Neuroscience*, *33*(1), 32–49. https://doi.org/10.1080/23273798.2017.1359634

Shapiro, D. N., Chandler, J., & Mueller, P. A. (2013). Using mechanical turk to study clinical populations. *Clinical Psychological Science*, *1*(2), 213–220.

Silberzahn, R., Uhlmann, E. L., Martin, D. P., Anselmi, P., Ullrich, J., Aust, F., … Bannard, C. (2017). Many analysts, one dataset: Making transparent how variations in analytical choices affect results. *Advances in Methods and Practices in Psychological Science*. https://doi.org/10.5167/uzh-148470

Simmons, J. P., Nelson, L. D., & Simonsohn, U. (2011). False-Positive Psychology: Undisclosed Flexibility in Data Collection and Analysis Allows Presenting Anything as Significant. *Psychological Science*, *22*(11), 1359–1366. https://doi.org/10.1177/0956797611417632

Simons, D. J., Holcombe, A. O., & Spellman, B. A. (2014). An introduction to registered replication reports at perspectives on psychological science. *Perspectives on Psychological Science*, *9*(5), 552–555. https://doi.org/10.1177/1745691614543974

Simonsohn, U. (2015). Small telescopes: Detectability and the evaluation of replication results. *Psychological Science*, *26*(5), 559–569.

Stack, C. M. H., James, A. N., & Watson, D. G. (2018). A failure to replicate rapid syntactic adaptation in comprehension. *Memory & Cognition*, *46*(6), 864–877. https://doi.org/10.3758/s13421-018-0808-6

Sterling, T. D. (1959). Publication decisions and their possible effects on inferences drawn from tests of significance–or vice versa. *Journal of the American Statistical Association*, *54*(285), 30–34. https://doi.org/10.1080/01621459.1959.10501497

Strack, F., Martin, L. L., & Stepper, S. (1988). Inhibiting and facilitating conditions of the human smile: A nonobtrusive test of the facial feedback hypothesis. *Journal of Personality and Social Psychology*, *54*(5), 768–777. https://doi.org/10.1037/0022-3514.54.5.768

Taylor, D. Q. (1975). The inadequacy of bipolarity and distinctive features: The German “voiced/voiceless” consonants. In *The Second LACUS Forum*(pp. 107–119).

Trubeckoj, N. S. (1939). *Grundzüge der Phonologie*(Vol. 7). Prag: Travaux du Cercle Linguistique de Prague.

Van Oostendorp, M. (2008). Incomplete devoicing in formal phonology. *Lingua*, *118*(9), 1362–1374.

Vasishth, S., Mertzen, D., Jäger, L. A., & Gelman, A. (2018). The statistical significance filter leads to overoptimistic expectations of replicability. *Journal of Memory and Language*, *103*, 151–175.

Vasishth, S., & Nicenboim, B. (2016). Statistical methods for linguistic research: Foundational ideas - Part I. *Language and Linguistics Compass*, *10*(8), 349–369. https://doi.org/10.1111/lnc3.12201

Wagenmakers, E.-J., Beek, T., Dijkhoff, L., Gronau, Q. F., Acosta, A., Adams, R. B., … Zwaan, R. A. (2016). Registered replication report: Strack, Martin, & Stepper (1988). *Perspectives on Psychological Science*, *11*(6), 917–928. https://doi.org/10.1177/1745691616674458

Wager, T. D., Lindquist, M. A., Nichols, T. E., Kober, H., & Van Snellenberg, J. X. (2009). Evaluating the consistency and specificity of neuroimaging data using meta-analysis. *Neuroimage*, *45*(1), S210–S221. https://doi.org/10.1016/j.neuroimage.2008.10.061

Wagner, P., Trouvain, J., & Zimmerer, F. (2015). In defense of stylistic diversity in speech research. *Journal of Phonetics*, *48*, 1–12. https://doi.org/10.1016/j.wocn.2014.11.001

Wickham, H. (2017). Tidyverse: Easily install and load ’tidyverse’ packages (Version 1.2.1). Retrieved from https://CRAN.R-project.org/package=tidyverse

Wiese, R. (1996). *The phonology of German*. Oxford: Clarendon Press.
Winter, B. (2011). Pseudoreplication in phonetic research. In *Proceedings of the International Congress of Phonetic Science*(pp. 2137–2140). Hong Kong.
Winter, B. (2015). The other N: The role of repetitions and items in the design of phonetic experiments. In *Proceedings of the 18th International Congress of Phonetic Sciences.*Glasgow: The University of Glasgow.
Winter, B., & Roettger, T. (2011). The nature of incomplete neutralization in German: Implications for laboratory phonology. *Grazer Linguistische Studien*, *76*, 55–74.
Xu, Y. (2010). In defense of lab speech. *Journal of Phonetics*, *38*(3), 329–336. https://doi.org/10.1016/j.wocn.2010.04.003
Zwaan, R. A., Etz, A., Lucas, R. E., & Donnellan, M. B. (2018). Making replication mainstream. *Behavioral and Brain Sciences*, *41*, E120. https://doi.org/10.1017/S0140525X17001972